Scene Classification Update 4

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Karina and Pauline:

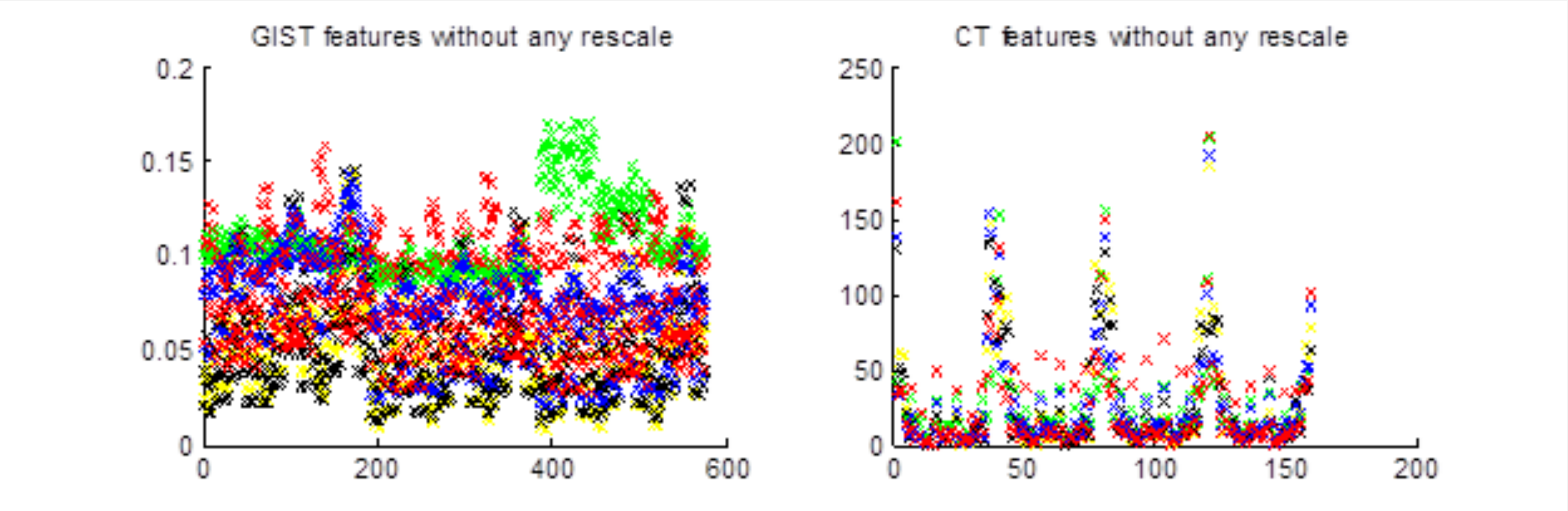
Gabor GIST + CENTRIST

Hybrid: Gabor GIST + CENTRIST

Visualize information of:

- GIST feature vectors
- CT feature vectors

Hybrid: Gabor GIST + CENTRIST



Very different scales!

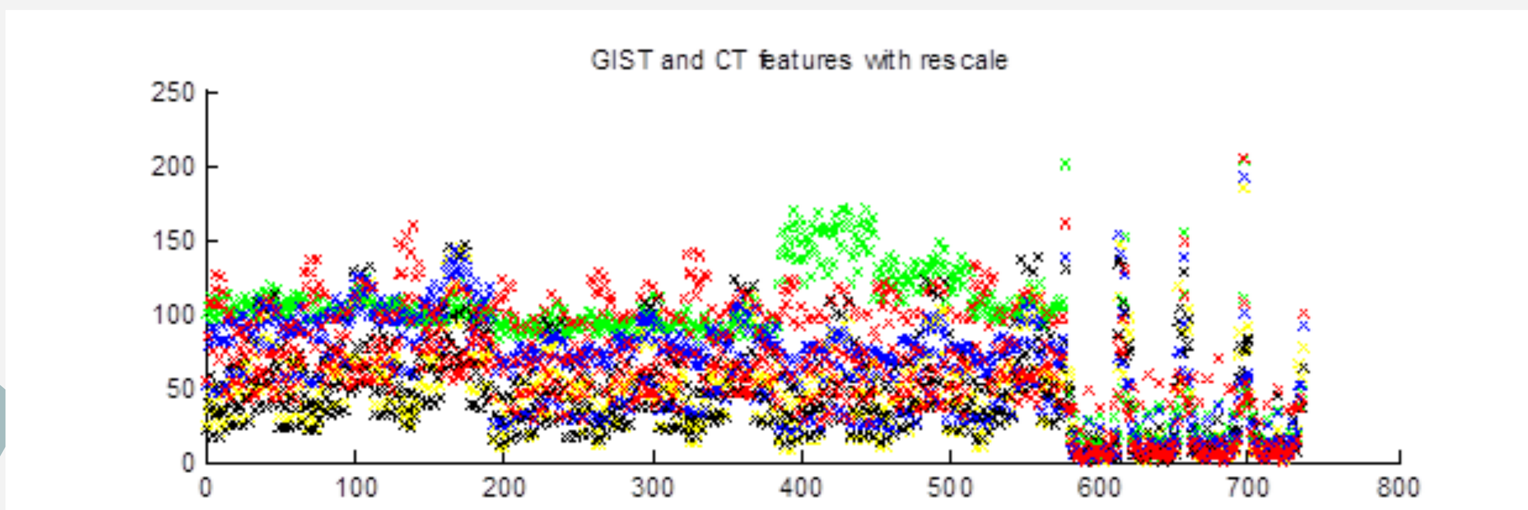
Hybrid: Gabor GIST + CENTRIST

How to scale?

Stretch Gabor GIST through CENTRIST scale

NN classifier rescales => absolute scale irrelevant
relative scale important

1. Find max and min of Centrist values: [a, b]
2. Scale Gabor into CENTRIST [a, b] range: A = (b-a)*Gabor + a



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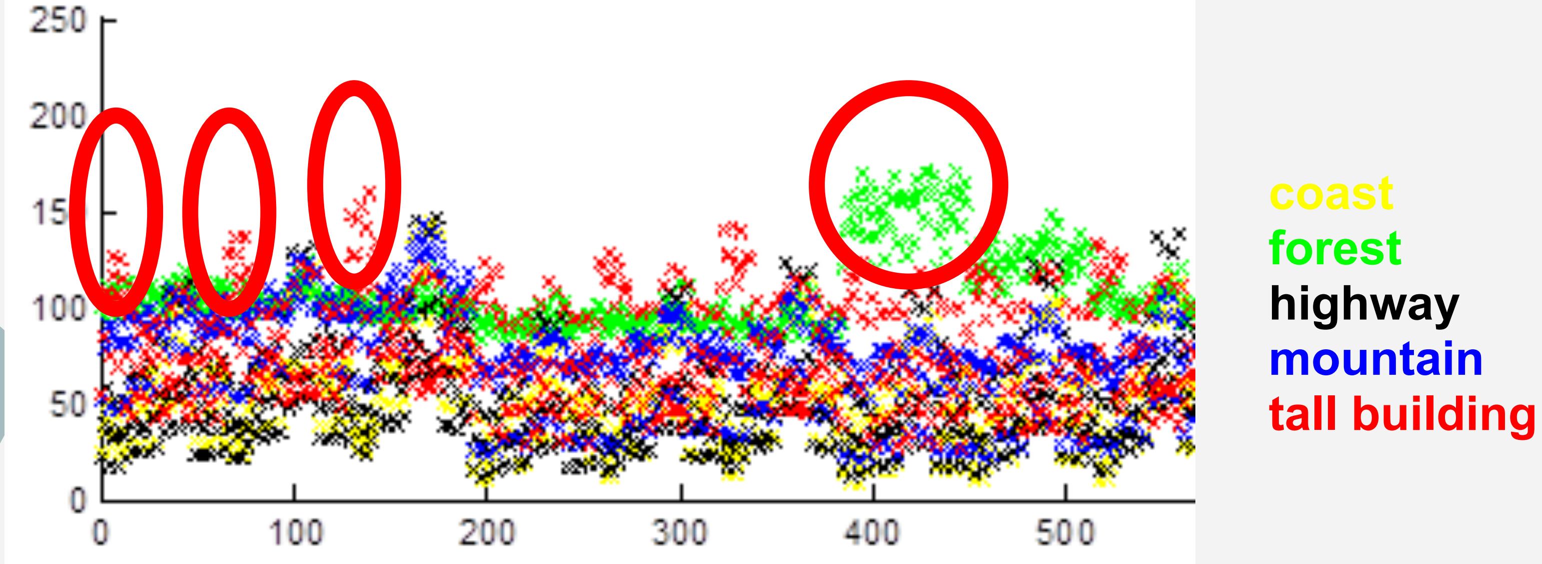
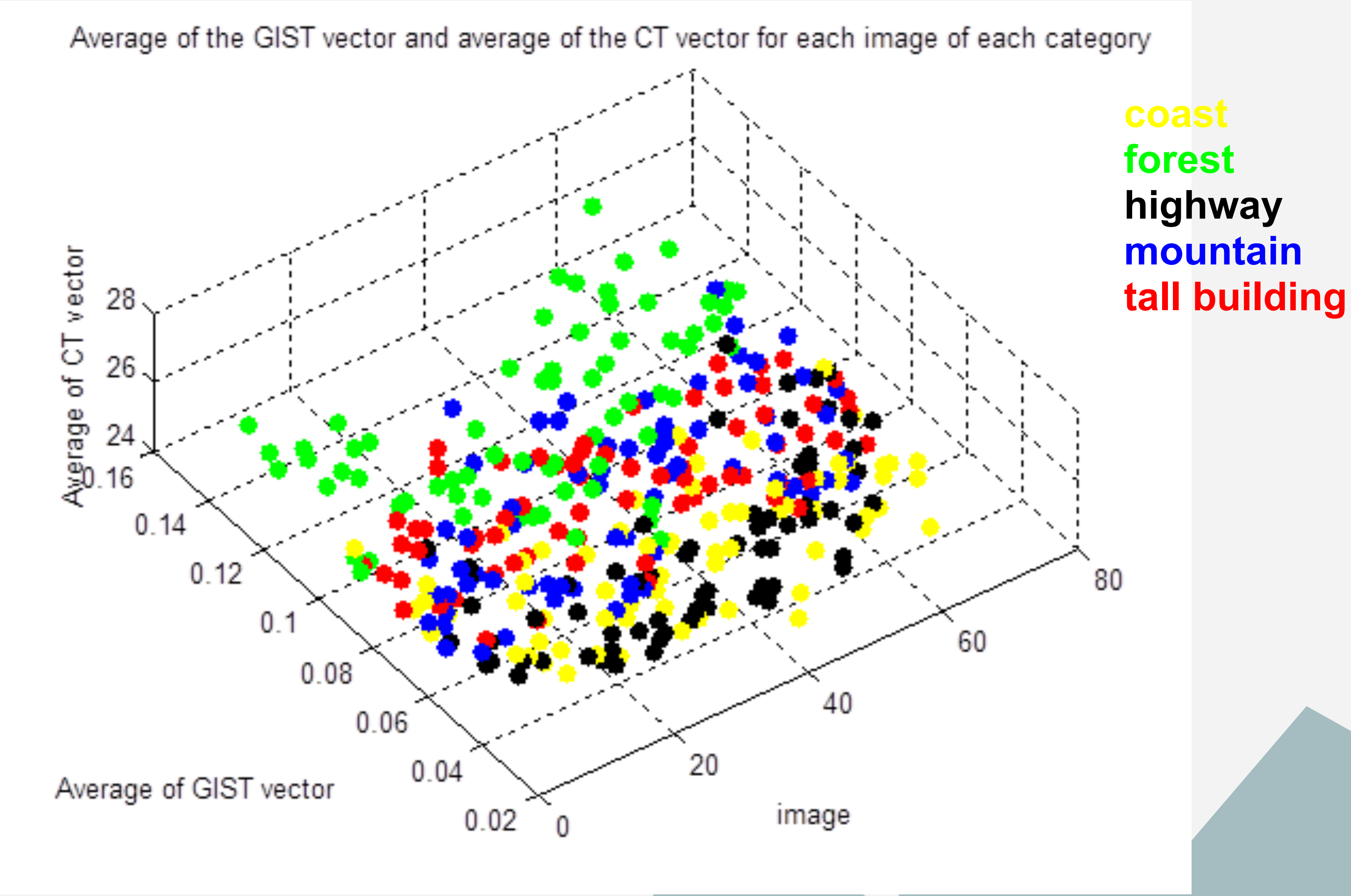
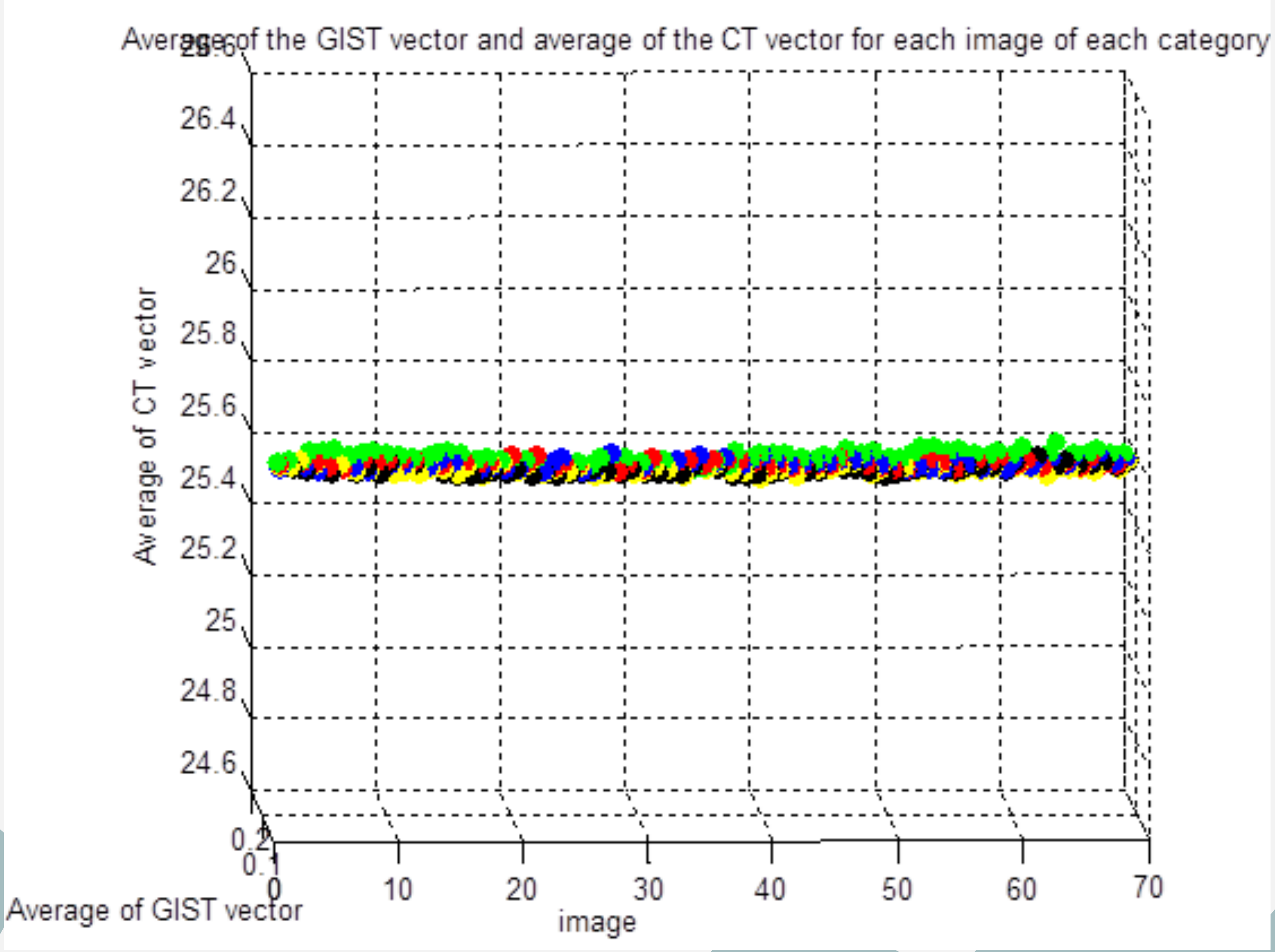
GIST feature vector:



Hybrid: Gabor GIST + CENTRIST

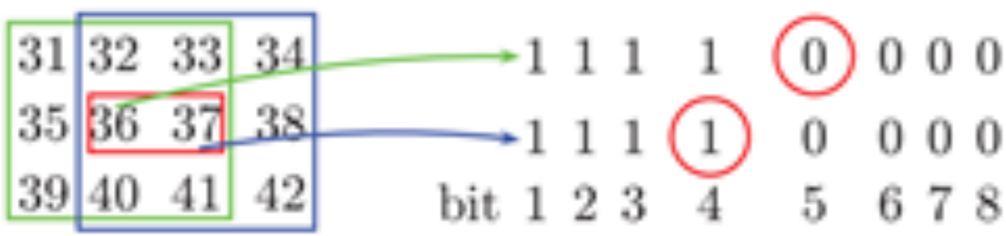


Hybrid: Gabor GIST + CENTRIST



CENTRIST descriptor

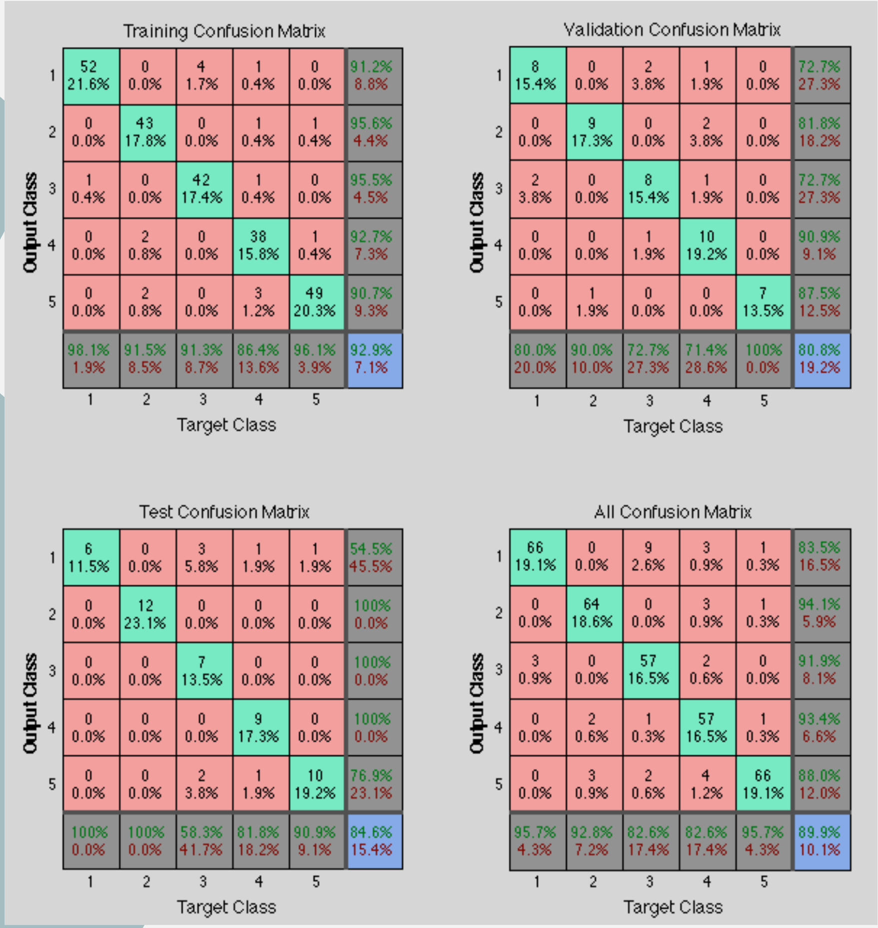
$$\begin{matrix} 32 & 64 & 96 \\ 32 & \mathbf{64} & 96 \\ 32 & 32 & 96 \end{matrix} \Rightarrow \begin{matrix} 1 & 1 & 0 \\ 1 & & 0 \\ 1 & 1 & 0 \end{matrix} \Rightarrow (11010110)_2 \Rightarrow CT = 214$$

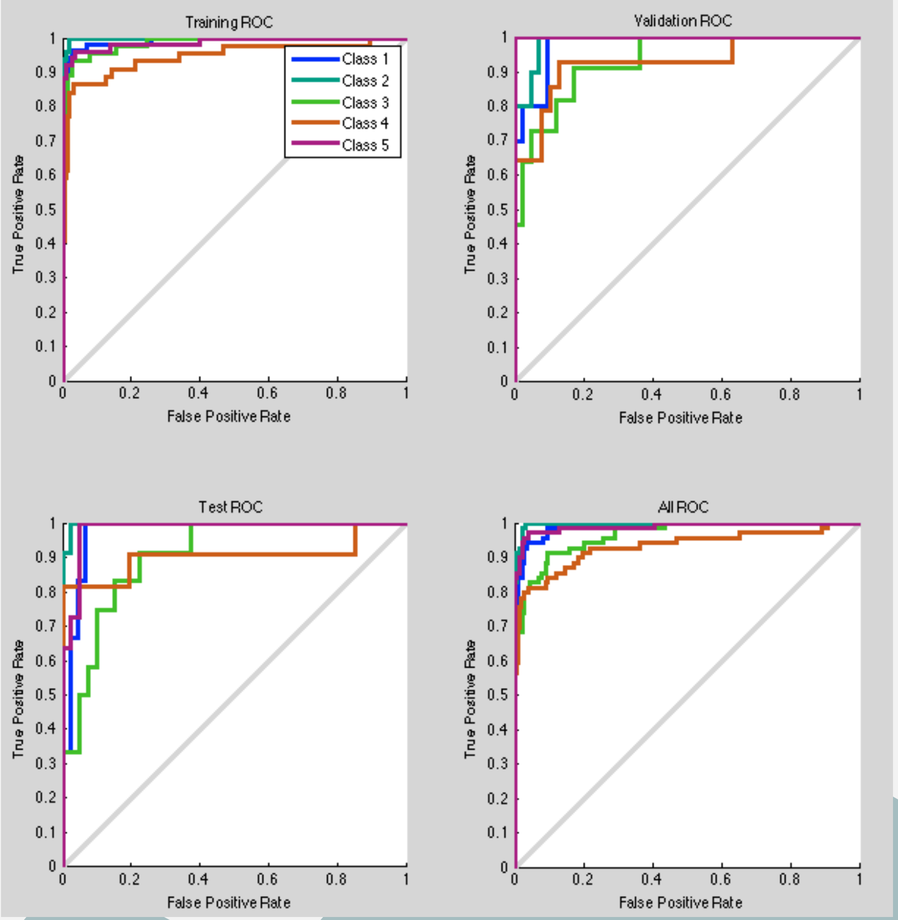


Constraints between neighbors => homogeneity introduced

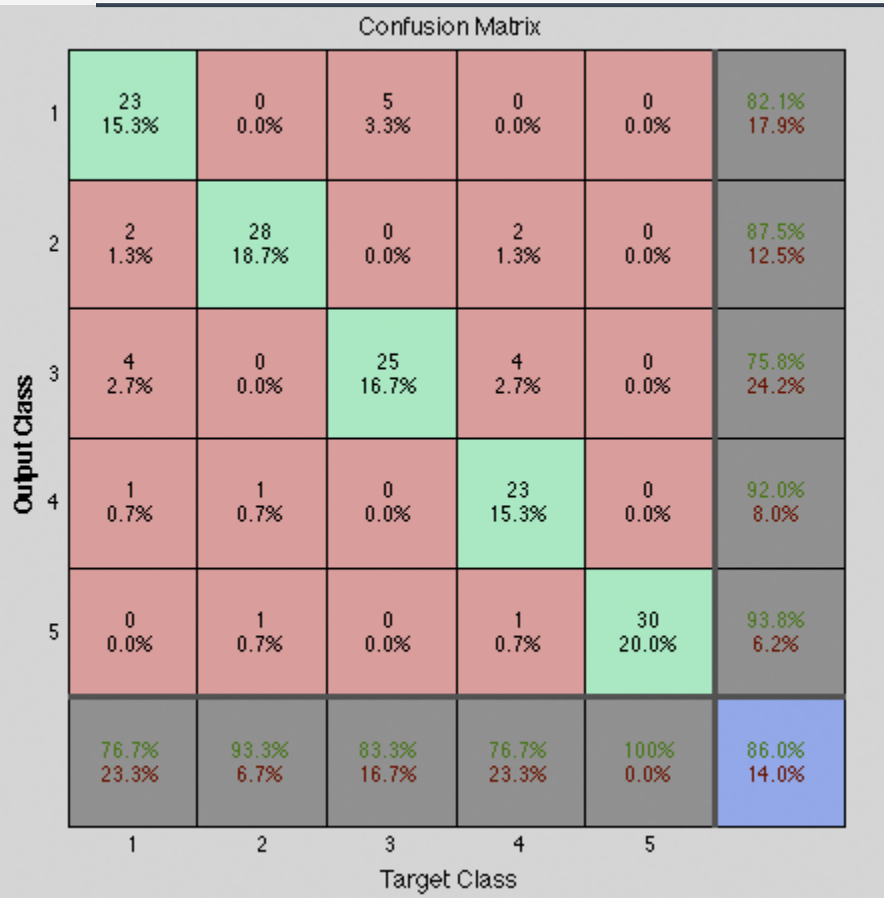
CENTRIST descriptor

Pure CENTRIST results through the network:



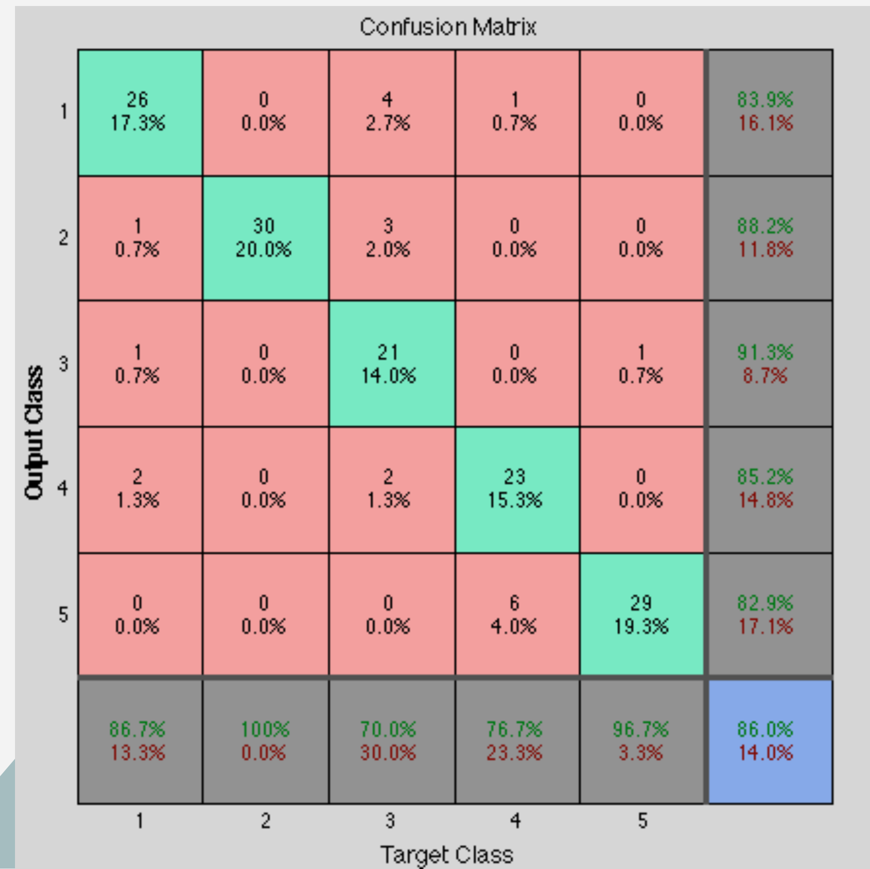


CENTRIST vs Gabor descriptor



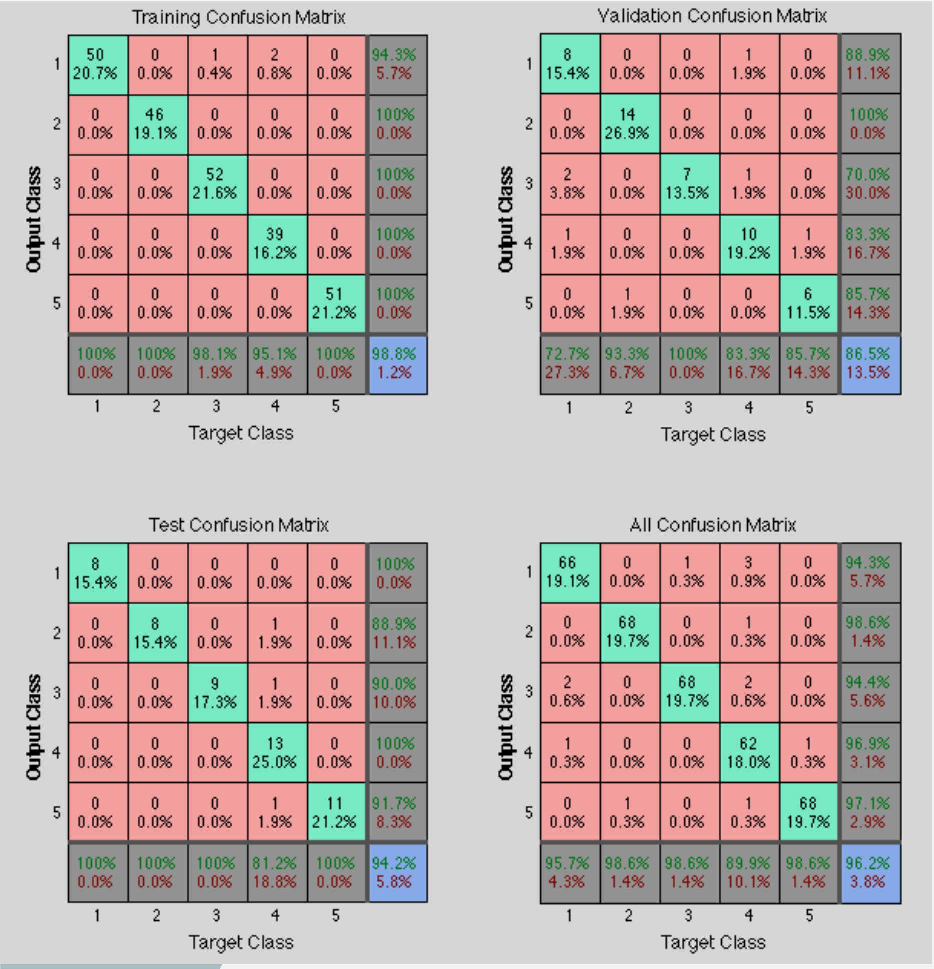
CENTRIST

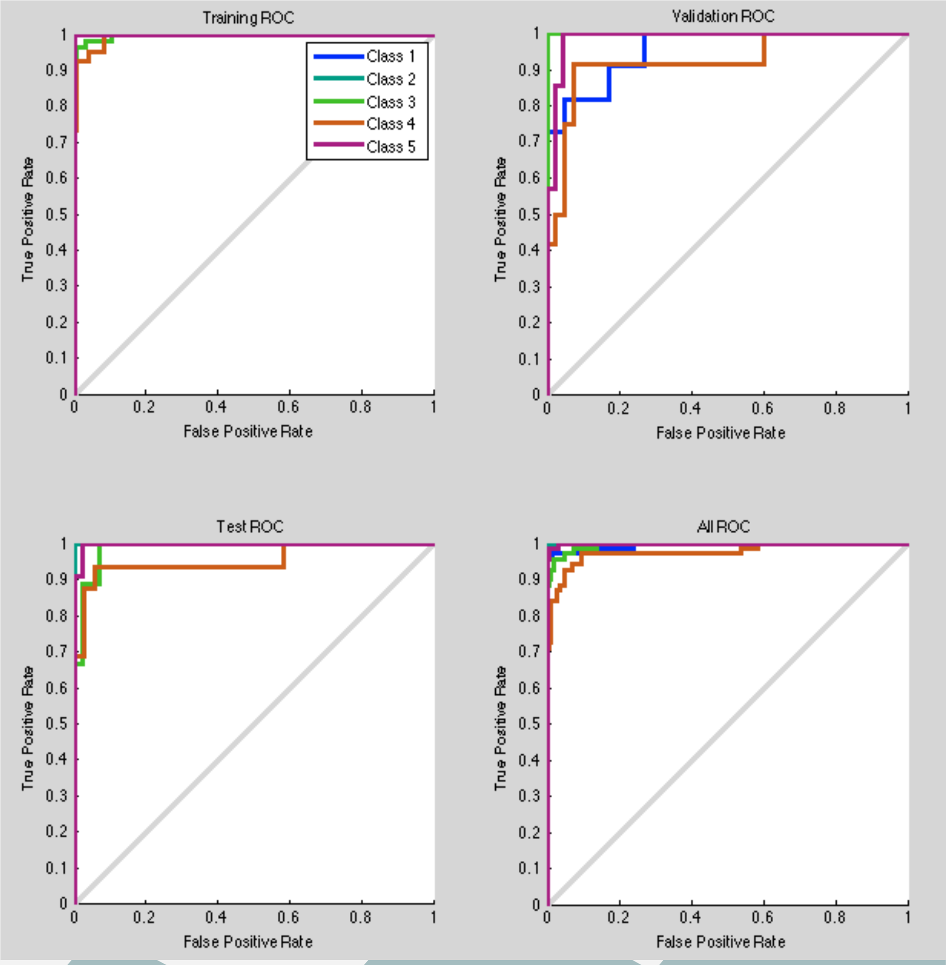
Gabor GIST



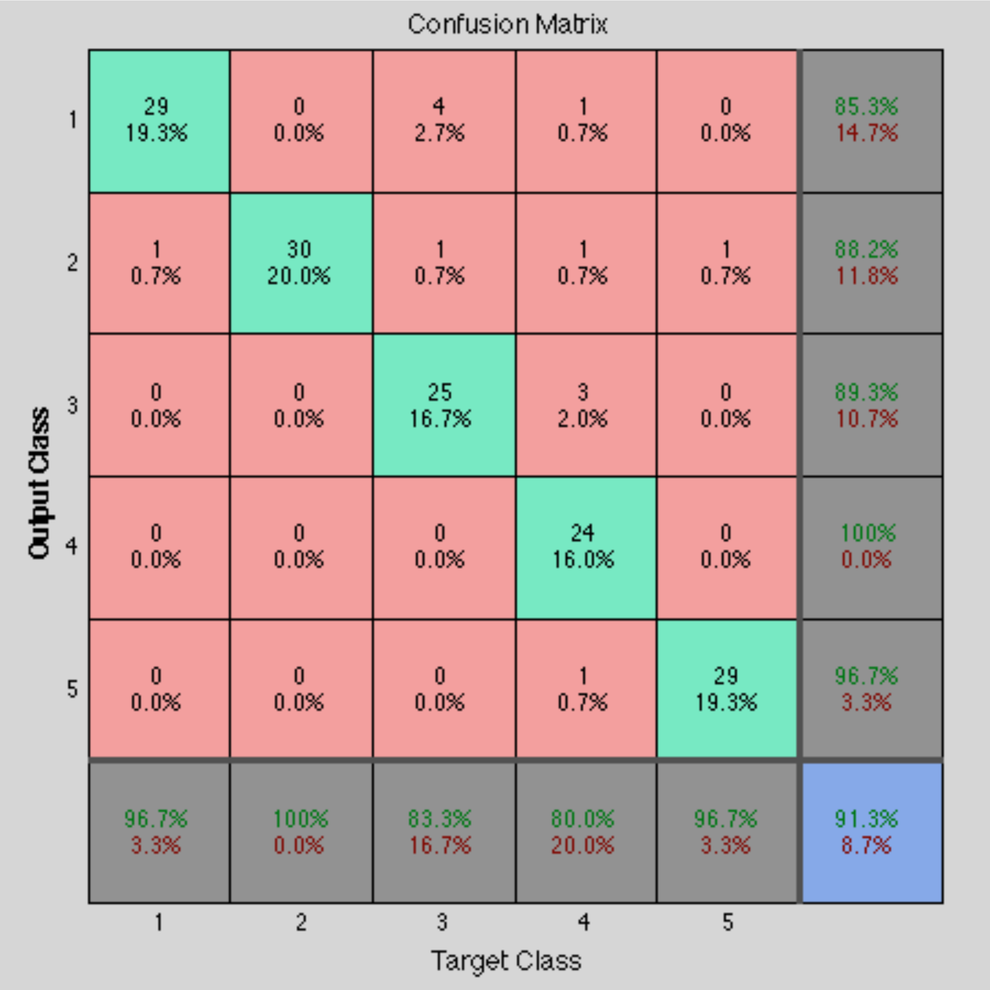
Hybrid: Gabor GIST + CENTRIST

736 inputs => 200 hidden nodes => 5 outputs





Hybrid: Gabor GIST + CENTRIST



91.3 % accuracy on 150 testing images dataset..

I have a feeling that it can be boost to 93 or so just by playing with number of hidden neurons and retrain it more and more

Coast and highway got improved comparably with Gabor

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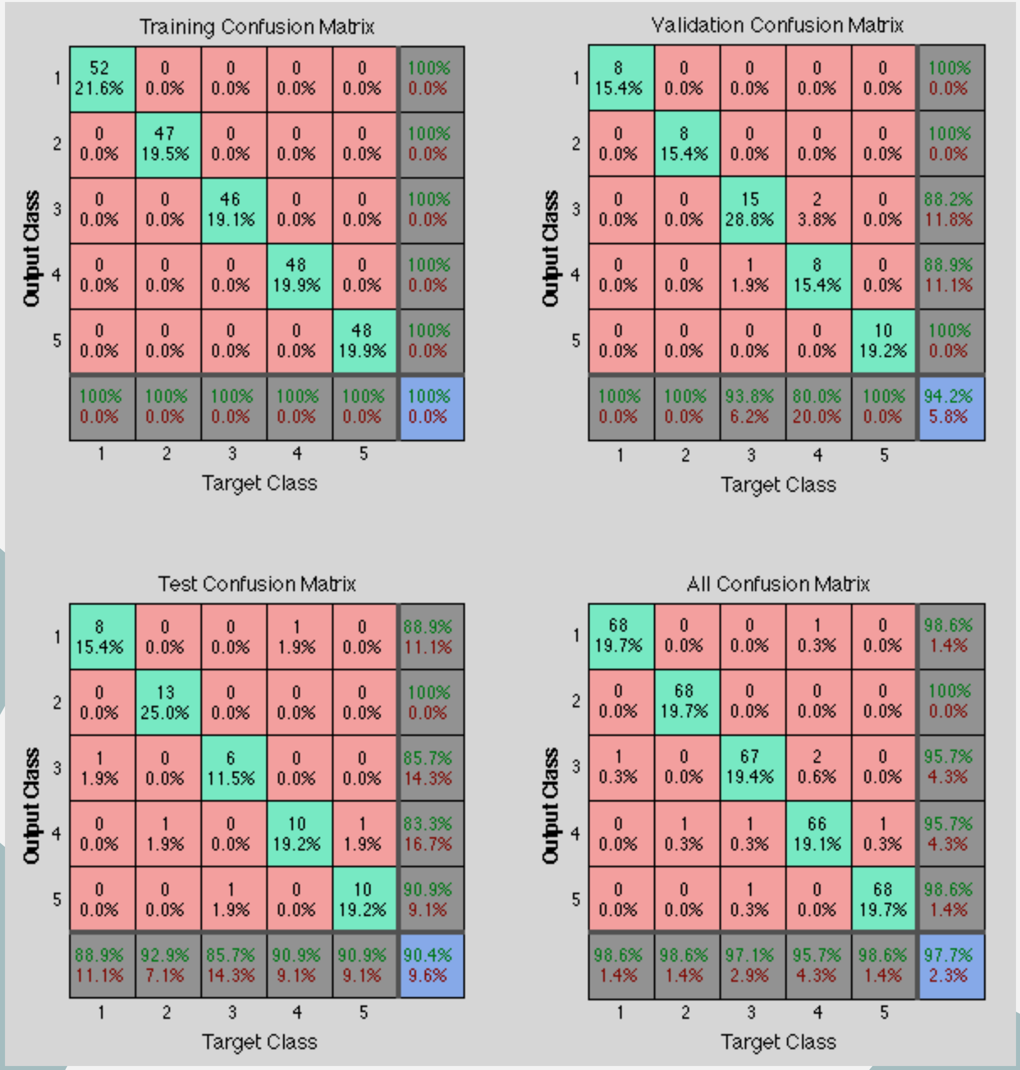

250 hidden nodes, worse training results, BUT turned to be better overall

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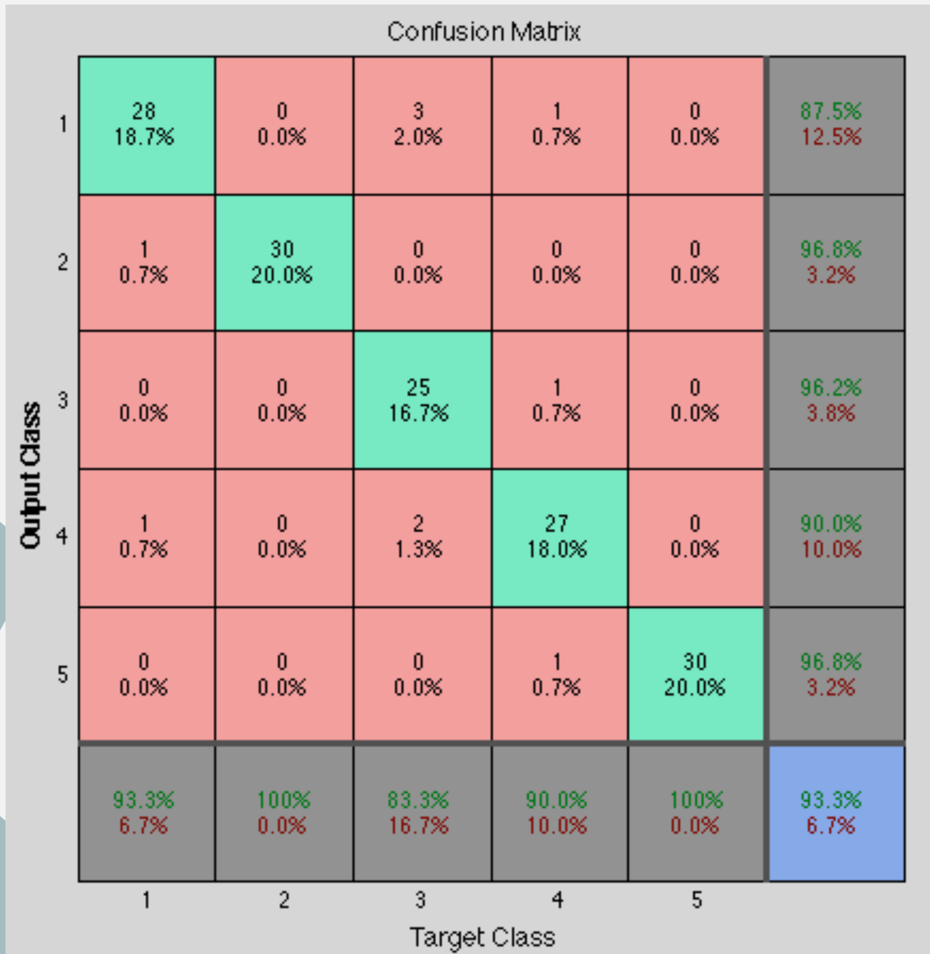


93.3 % accuracy on 150 testing images dataset

Improved Mountain and Tall building, but misclassified one coast comparably with 200 nodes.

Now: 100% accuracy for Forest and Tall building

Gabor descriptor (addition)

About whitening:

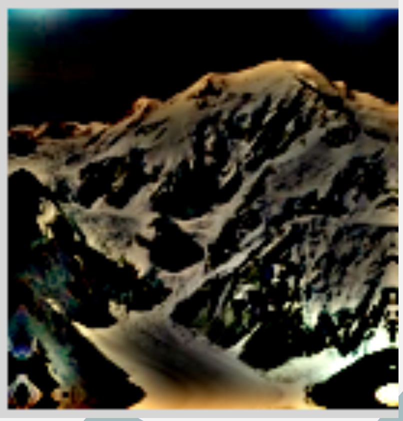
Purpose: whitening is simply a process for removing *pairwise*, *linear* correlations => remove noisy details

How: through designing a filter in the frequency domain that will flatten the spectrum of a natural image (on average)

- design the amplitude spectrum of the filter so that it rises linearly with frequency, to compensate for the $1/f$ amplitude spectrum of natural images [define a grid of frequency coordinate]

- convolution in the space domain = multiplication in the frequency domain, => multiply the spectrum of the image with the spectrum of the filter and take the inverse fft2

Just check this if interested:
http://redwood.berkeley.edu/bruno/npb261b/lab2/lab2.html



Piter:

Color histogram descriptor

COLOR HISTOGRAM descriptor